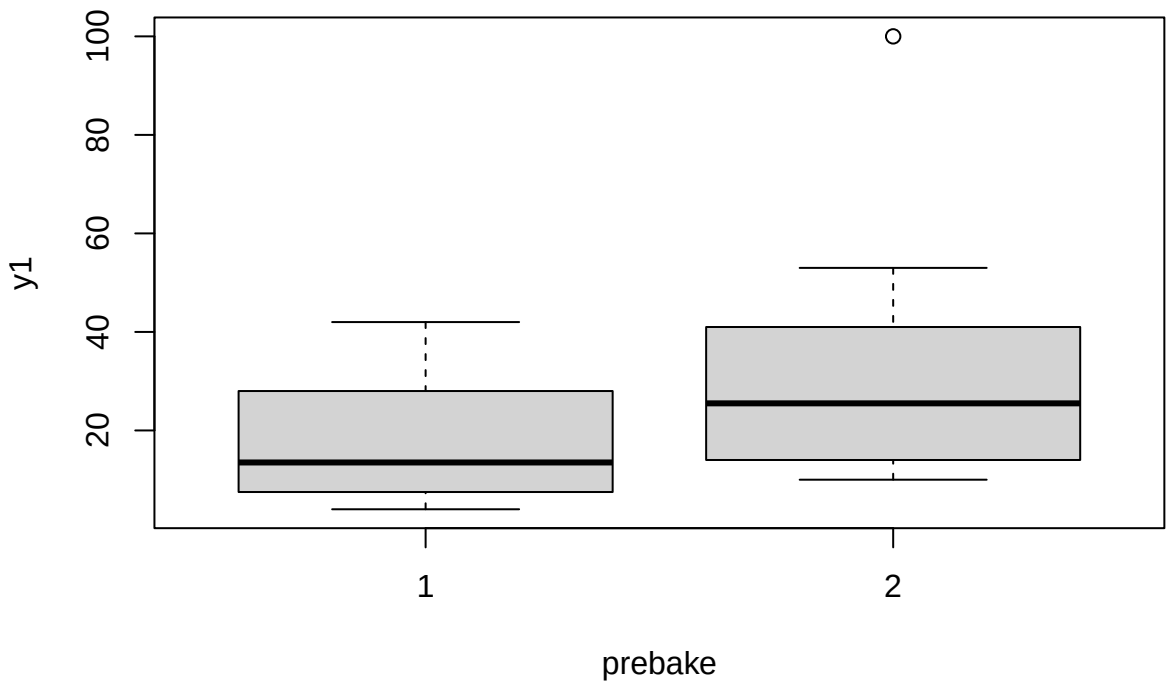
STOR 557 HW 5

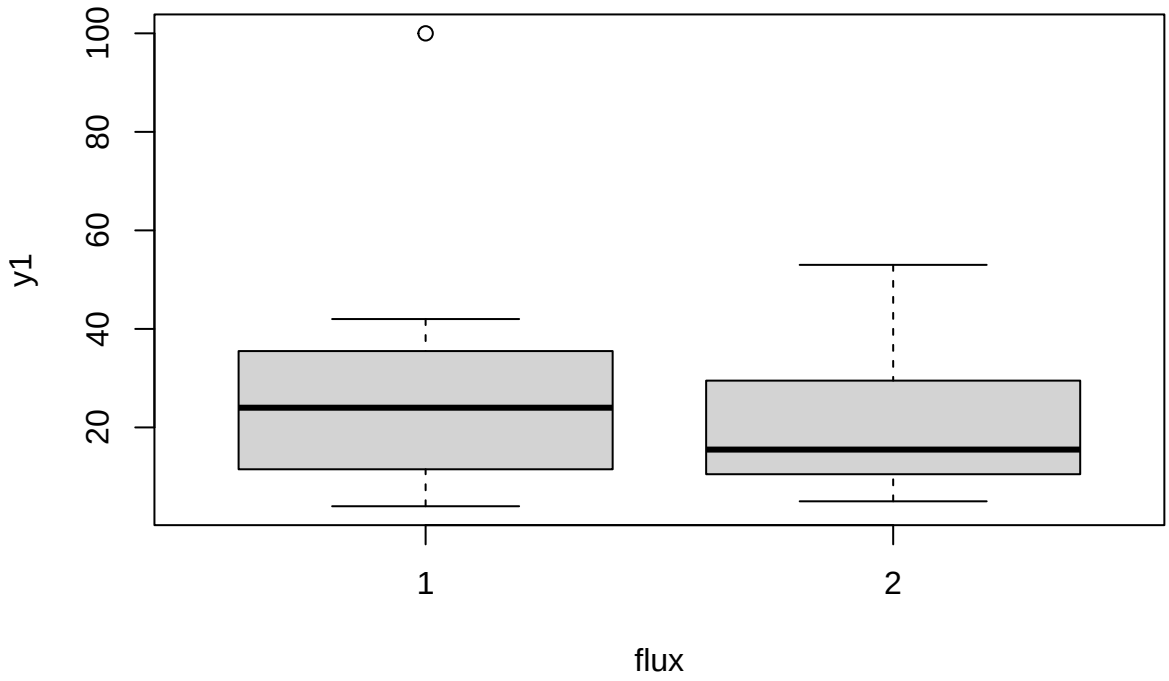
Branda Sisoutham

2026-10-01

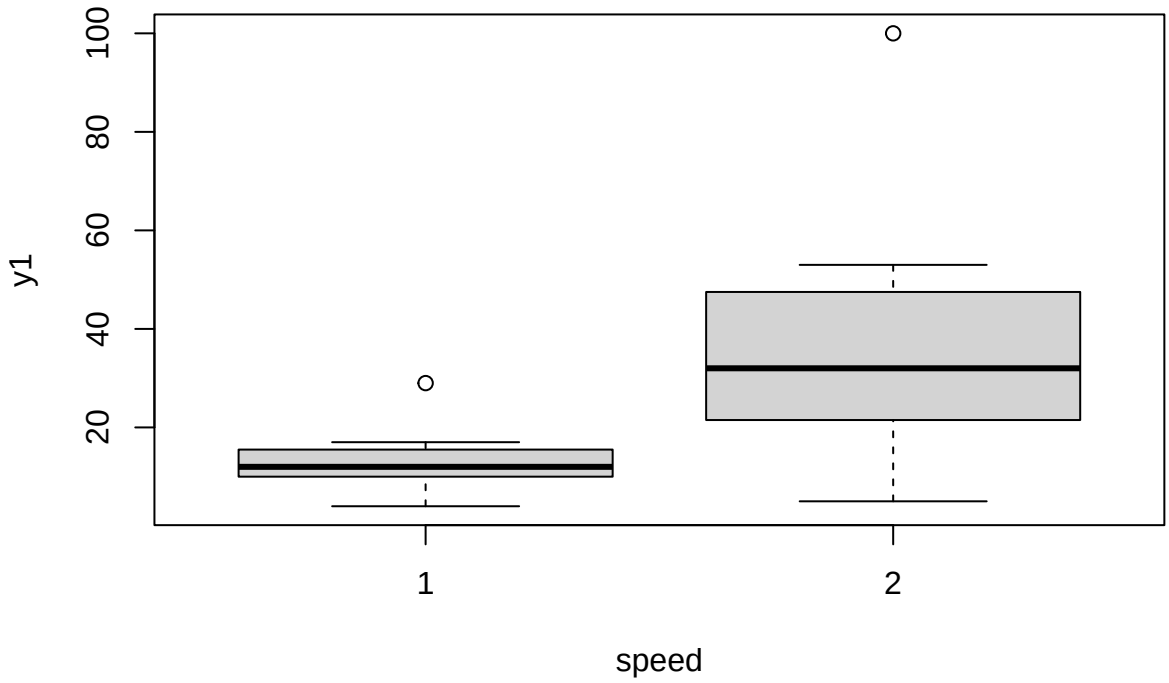
```
plot(y1 ~ prebake, data = wavesolder)
```



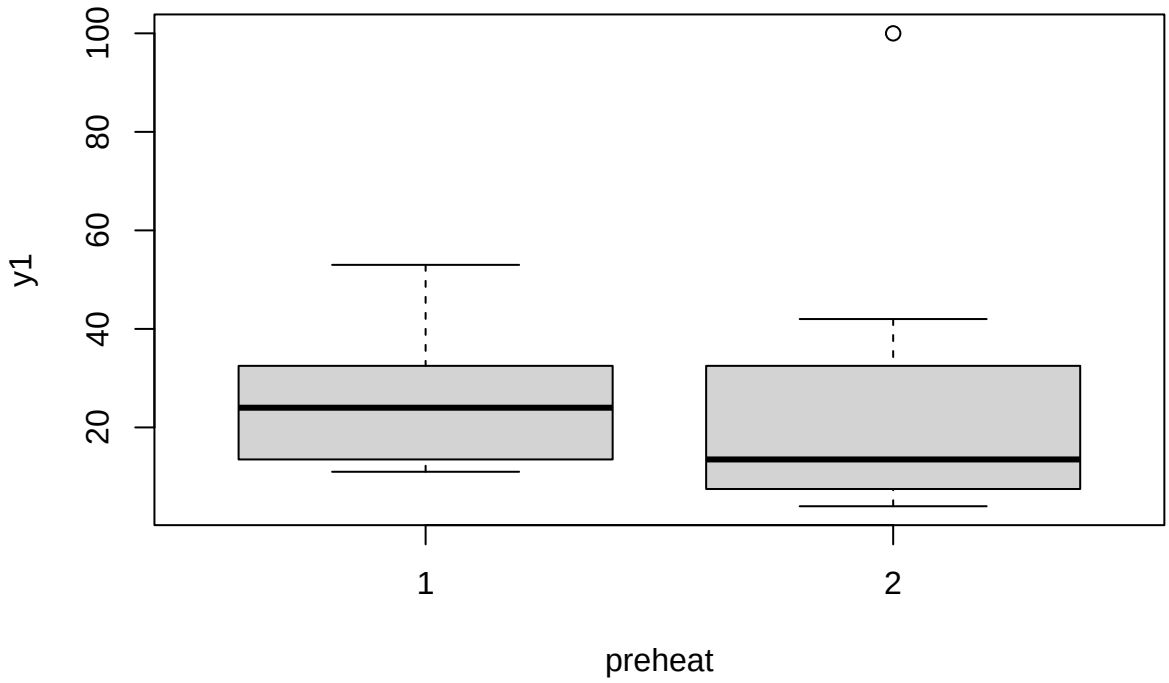

```
plot(y1 ~ flux, data = wavesolder)
```



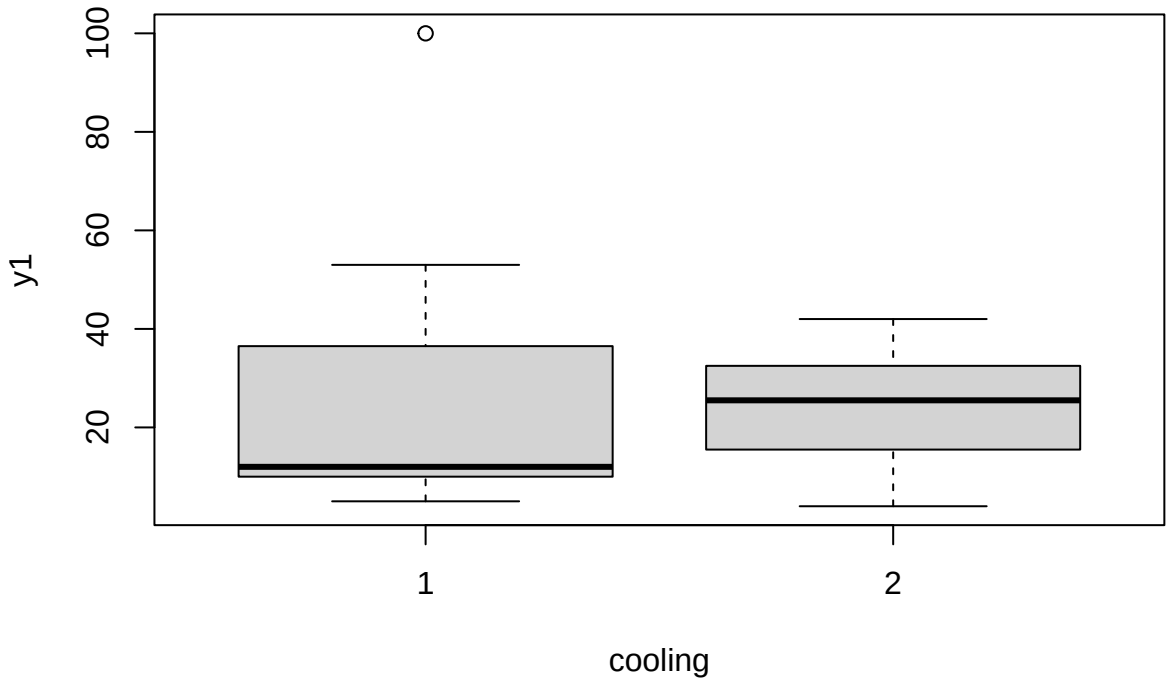

```
plot(y1 ~ speed, data = wavesolder)
```



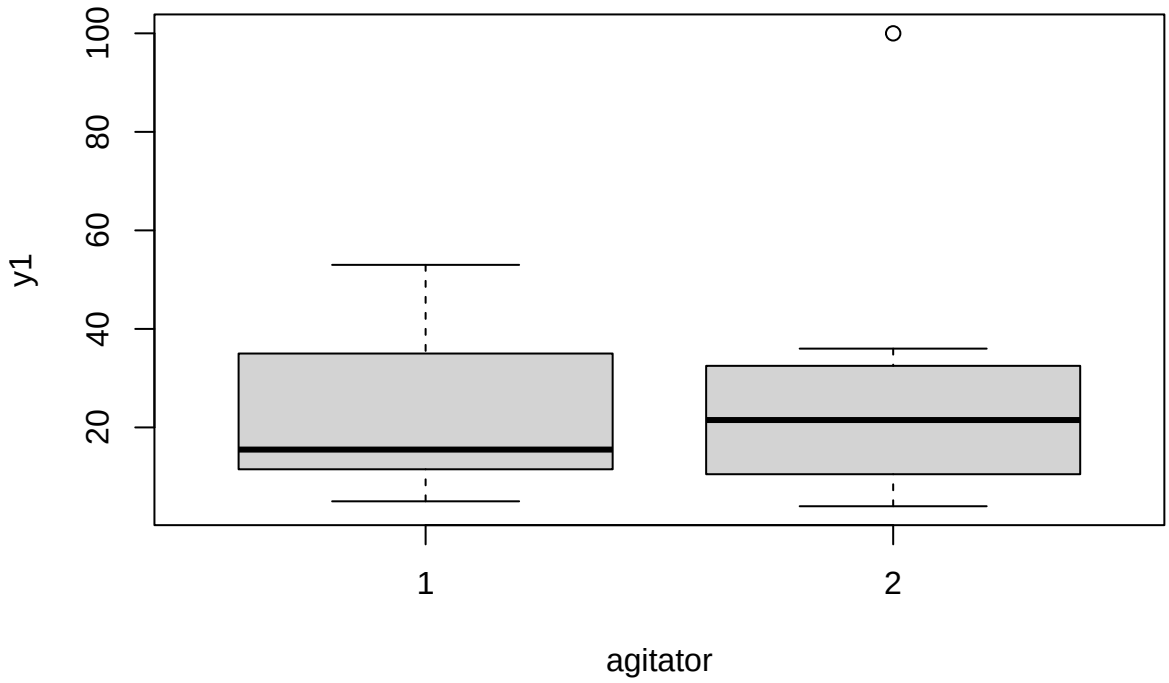

```
plot(y1 ~ preheat, data = wavesolder)
```



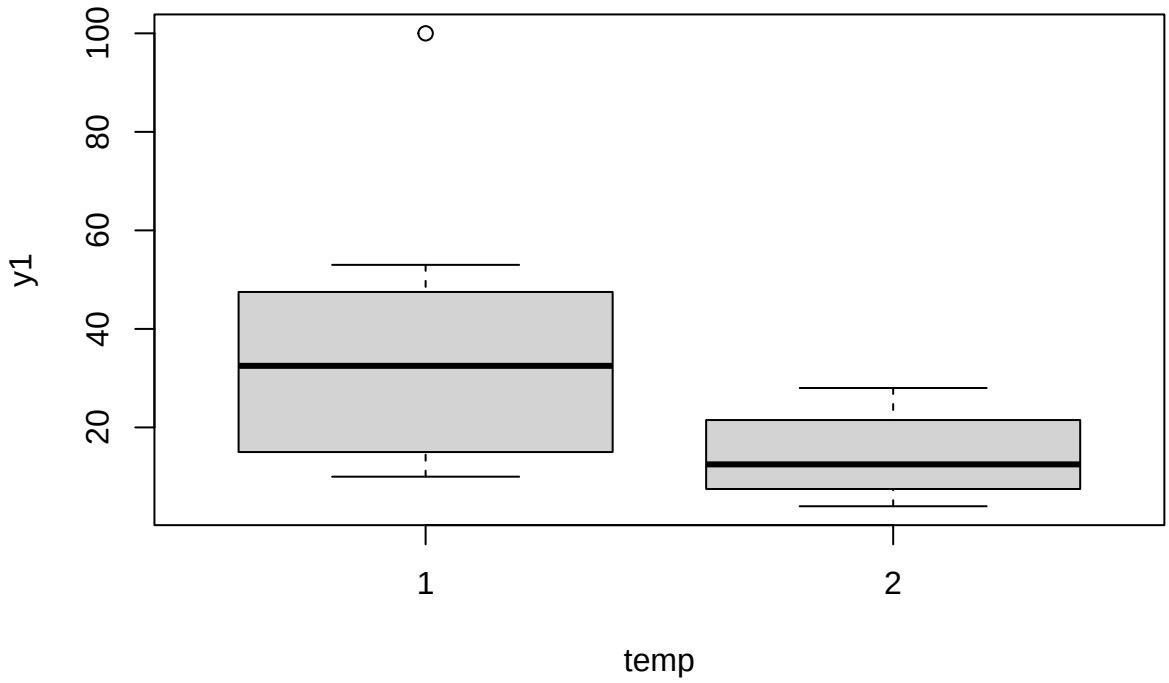

```
plot(y1 ~ cooling, data = wavesolder)
```



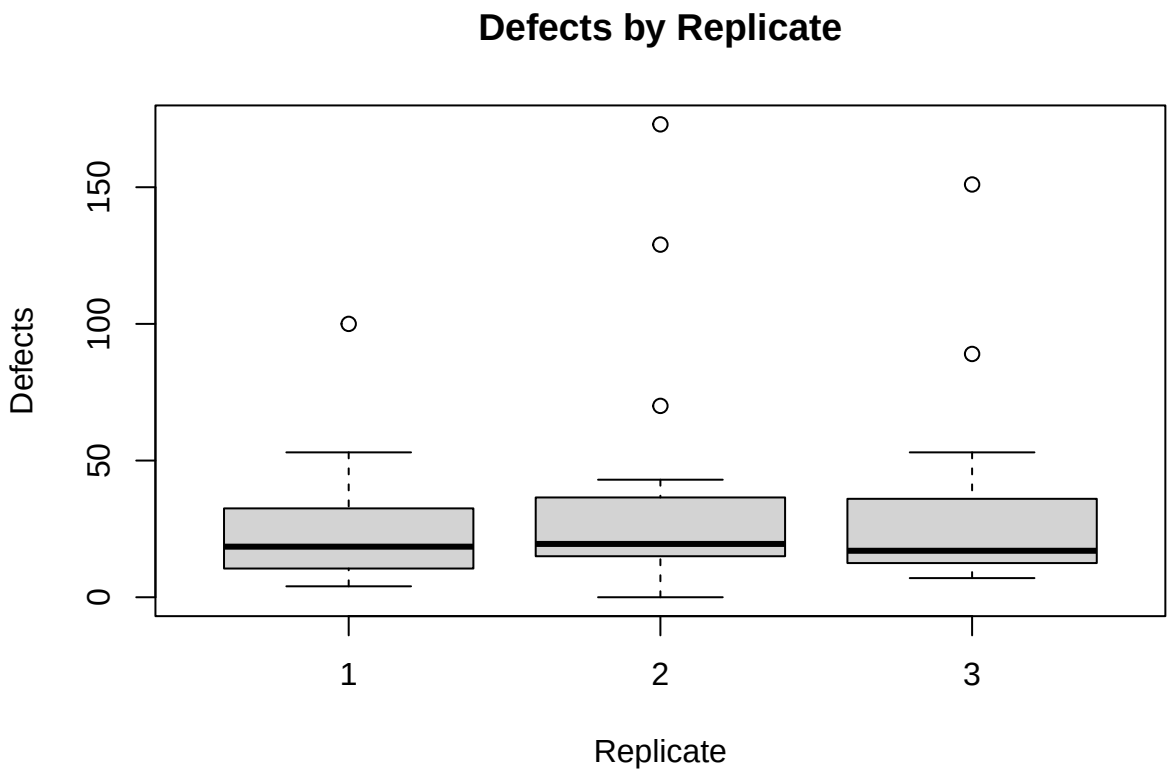

```
plot(y1 ~ agitator, data = wavesolder)
```




```
plot(y1 ~ temp, data = wavesolder)
```




```
boxplot(wavesolder$y1, wavesolder$y2, wavesolder$y3,
        names = c("1", "2", "3"),
        xlab = "Replicate",
        ylab = "Defects",
        main = "Defects by Replicate")
```



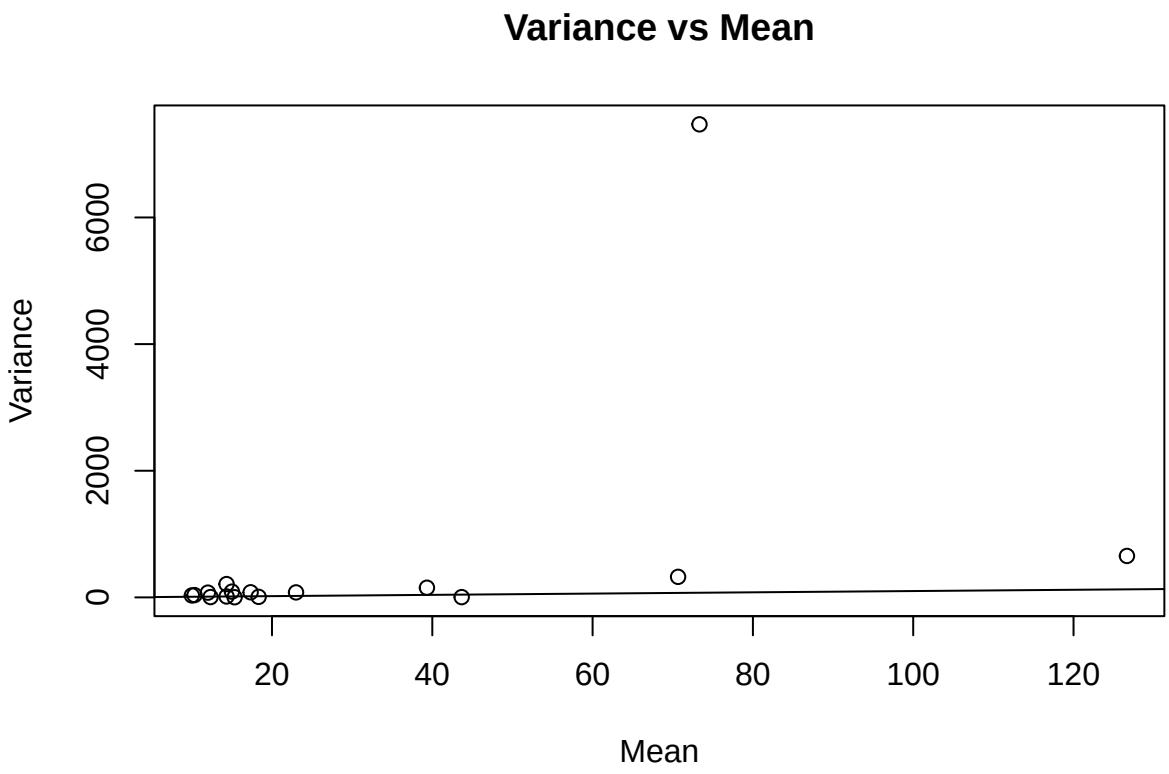
Speed and temperature seem to affect defects the most. The three replicates look similar, so there is no clear trend.

```
m <- rowMeans(wavesolder[, c("y1", "y2", "y3")])

v <- apply(wavesolder[, c("y1", "y2", "y3")], 1, var)

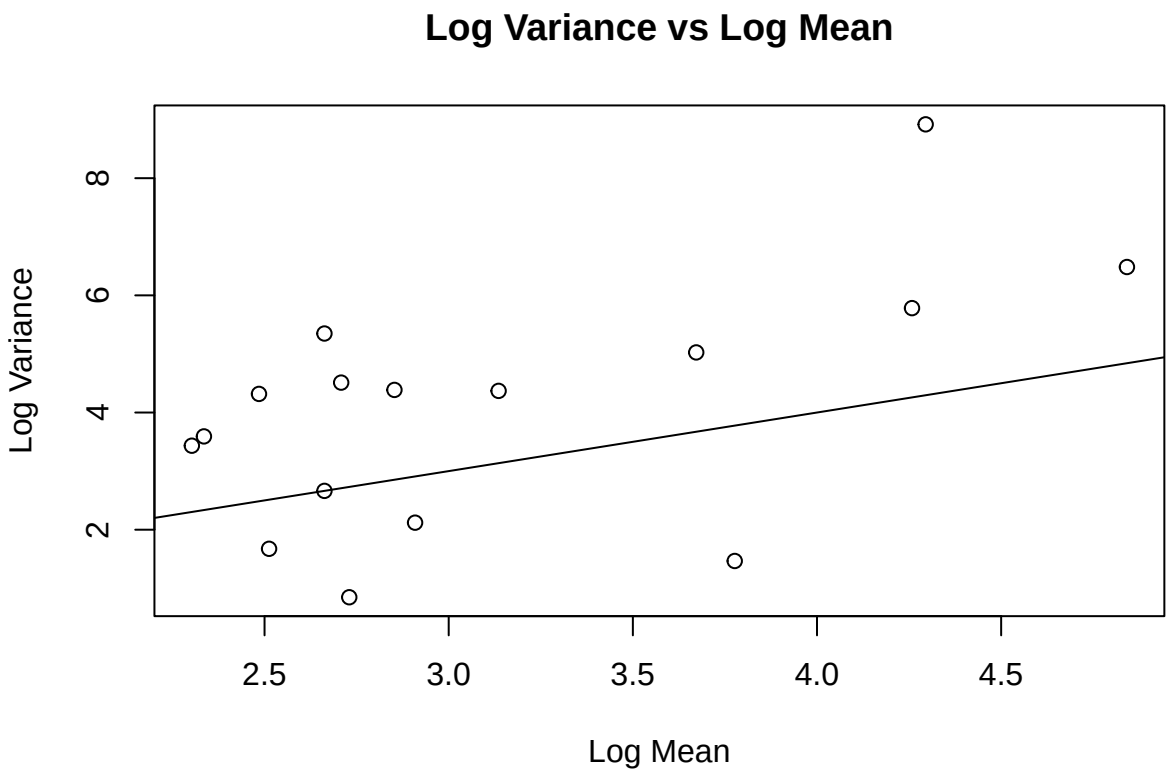
plot(m, v,
     xlab = "Mean",
     ylab = "Variance",
     main = "Variance vs Mean")

abline(0, 1)
```




```
plot(log(m), log(v),
     xlab = "Log Mean",
     ylab = "Log Variance",
     main = "Log Variance vs Log Mean")

abline(0, 1)
```



The variance is usually much larger than the mean, so the Poisson model may not fit well. The log plot also shows that the variance increases as the mean increases.

```
ws <- data.frame(
  defects = c(wavesolder$y1, wavesolder$y2, wavesolder$y3),
  replicate = factor(rep(1:3, each = nrow(wavesolder))),
  prebake = rep(wavesolder$prebake, 3),
  flux = rep(wavesolder$flux, 3),
  speed = rep(wavesolder$speed, 3),
  preheat = rep(wavesolder$preheat, 3),
  cooling = rep(wavesolder$cooling, 3),
  agitator = rep(wavesolder$agitator, 3),
  temp = rep(wavesolder$temp, 3)
)
```

```
mod1 <- glm(defects ~ prebake + flux + speed + preheat +
              cooling + agitator + temp,
            family = poisson,
            data = ws)

summary(mod1)
```

```
##
## Call:
## glm(formula = defects ~ prebake + flux + speed + preheat + cooling +
##     agitator + temp, family = poisson, data = ws)
```

```
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.04282    0.08560  35.548  < 2e-16 ***
## prebake2     0.65124    0.05442  11.968  < 2e-16 ***
## flux2       -0.57667    0.05438 -10.604  < 2e-16 ***
## speed2       1.22875    0.06113  20.100  < 2e-16 ***
## preheat2    -0.16645    0.05343  -3.115  0.00184 **
## cooling2    -0.16187    0.05313  -3.047  0.00231 **
## agitator2   -0.08974    0.05263  -1.705  0.08819 .
## temp2       -0.69816    0.05537 -12.609  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 1450.52  on 47  degrees of freedom
## Residual deviance:  491.78  on 40  degrees of freedom
## AIC: 738.52
##
## Number of Fisher Scoring iterations: 5
```

```
deviance(mod1)
```

```
## [1] 491.7759
```

```
df.residual(mod1)
```

```
## [1] 40
```

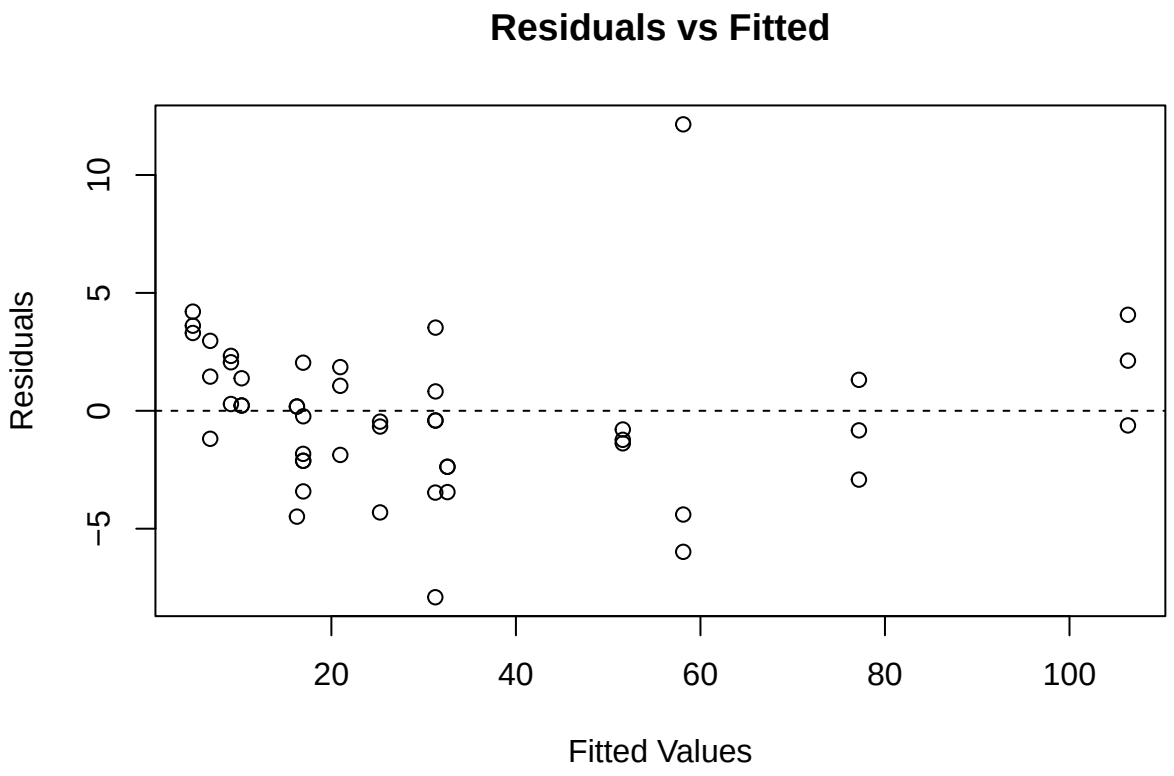
```
pchisq(deviance(mod1),
       df.residual(mod1),
       lower.tail = FALSE)
```

```
## [1] 3.853838e-79
```

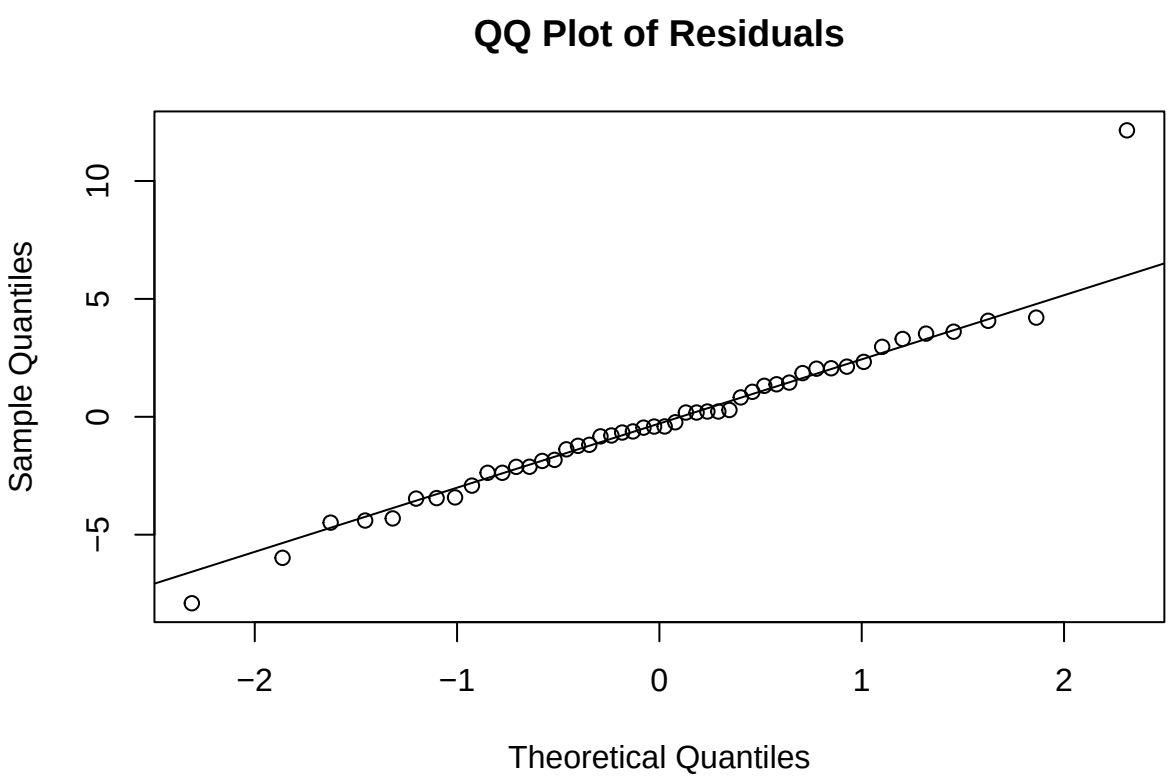
The Poisson model does not fit well because the residual deviance 491.8 is much larger than the degrees of freedom 40, and the p-value is extremely small.

```
plot(fitted(mod1), residuals(mod1),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted")

abline(h = 0, lty = 2)
```




```
qqnorm(residuals(mod1),
       main = "QQ Plot of Residuals")
qqline(residuals(mod1))
```



The residual plot shows some spread, and the QQ plot shows one clear outlier with a very large positive residual.

```
i <- which.max(abs(residuals(mod1)))

i
```

```
## 27
## 27
```

```
ws[i, ]
```

```
##    defects replicate prebake flux speed preheat cooling agitator temp
## 27     173         2       2    1     2       1       2        1    2
```

```
mod1e <- glm(defects ~ prebake + flux + speed + preheat +
               cooling + agitator + temp,
             family = poisson,
             data = ws[-i, ])

deviance(mod1e)
```

```
## [1] 251.7398
```

```r
df.residual(mod1e)
```

```
## [1] 39
```

```r
pchisq(deviance(mod1e),
       df.residual(mod1e),
       lower.tail = FALSE)
```

```
## [1] 6.448303e-33
```

After removing the outlier, the residual deviance dropped to 251.7 with 39 degrees of freedom, but the p-value is still extremely small. So the Poisson model still does not fit the data well.

```r
mod1f <- glm(defects ~ prebake + flux + speed + preheat +
               cooling + agitator + temp,
             family = quasipoisson,
             data = ws[-i, ])

summary(mod1f)
```

```
##
## Call:
## glm(formula = defects ~ prebake + flux + speed + preheat + cooling +
##     agitator + temp, family = quasipoisson, data = ws[-i, ])
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.11004    0.22504  13.820  < 2e-16 ***
## prebake2     0.42013    0.14909   2.818  0.00755 **
## flux2       -0.33593    0.14897  -2.255  0.02982 *
## speed2       1.05490    0.16134   6.538 9.28e-08 ***
## preheat2     0.03288    0.14702   0.224  0.82420
## cooling2    -0.39813    0.14930  -2.667  0.01109 *
## agitator2    0.10100    0.14454   0.699  0.48885
## temp2       -0.98669    0.16216  -6.085 3.95e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 6.641214)
##
##     Null deviance: 1137.32  on 46  degrees of freedom
## Residual deviance:  251.74  on 39  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

```r
summary(mod1f)$dispersion
```

```
## [1] 6.641214
```

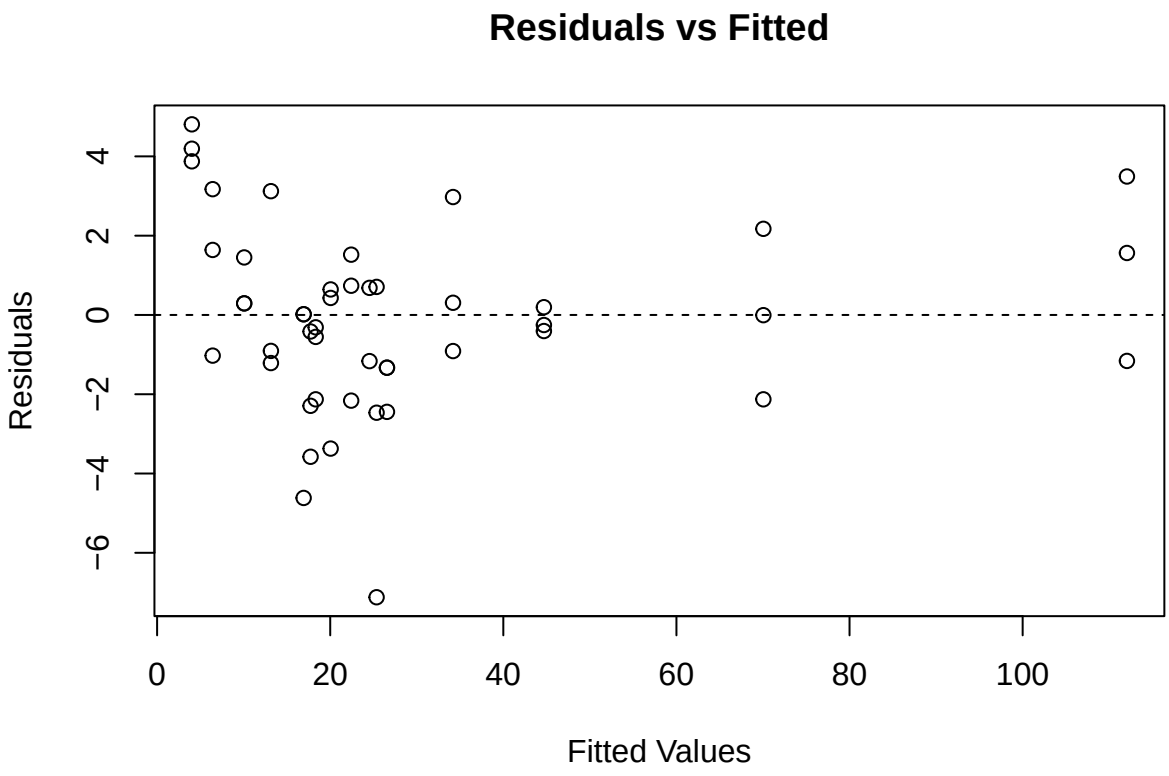
```r
drop1(mod1f, test = "F")
```

```
## Single term deletions
##
## Model:
## defects ~ prebake + flux + speed + preheat + cooling + agitator +
##     temp
##          Df Deviance F value    Pr(>F)
## <none>       251.74
## prebake   1   305.33  8.3029  0.006404 **
## flux      1   285.87  5.2879  0.026914 *
## speed     1   574.23 49.9600 1.728e-08 ***
## preheat   1   252.07  0.0515  0.821701
## cooling   1   299.97  7.4719  0.009373 **
## agitator  1   254.99  0.5030  0.482398
## temp      1   530.75 43.2249 8.267e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

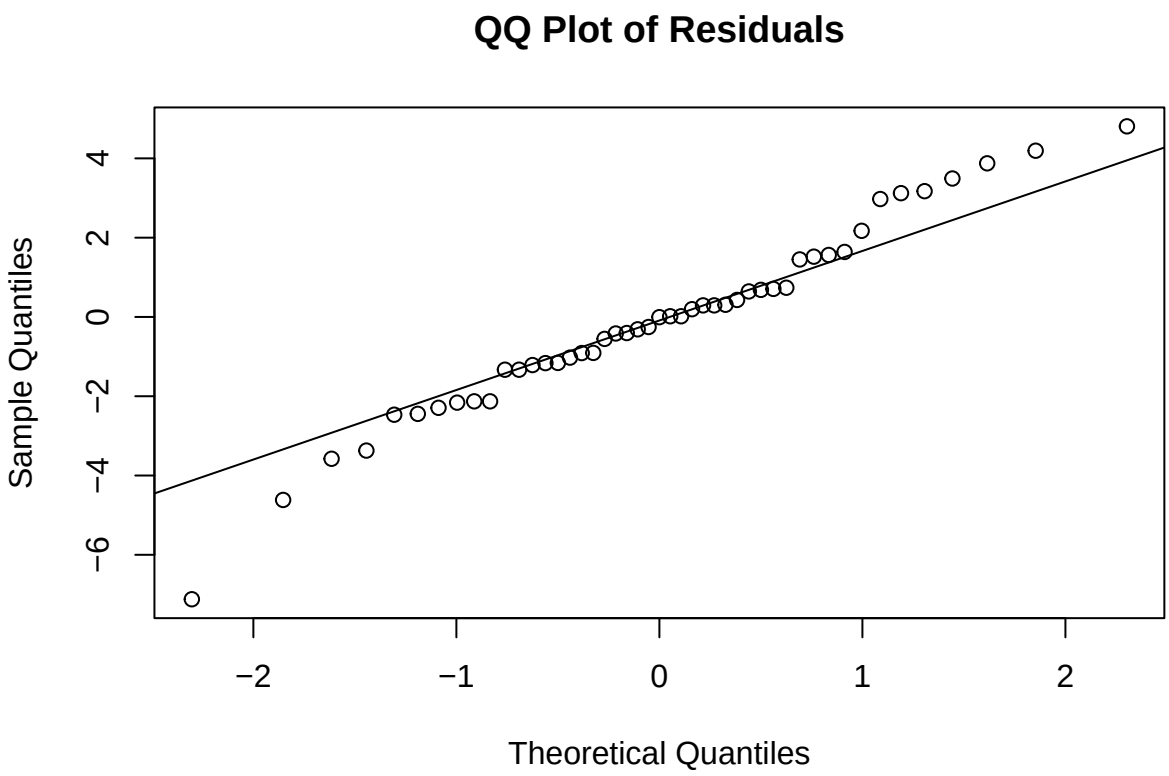
The dispersion parameter is 6.64, showing overdispersion. The model summary and F-tests give similar results. Prebake, flux, speed, cooling, and temperature are significant, while preheat and agitator are not. More defects are associated with prebake level 2, speed level 2, flux level 1, cooling level 1, and temperature level 1.

```r
plot(fitted(mod1f), residuals(mod1f),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted")

abline(h = 0, lty = 2)
```




```
qqnorm(residuals(mod1f),
       main = "QQ Plot of Residuals")
qqline(residuals(mod1f))
```



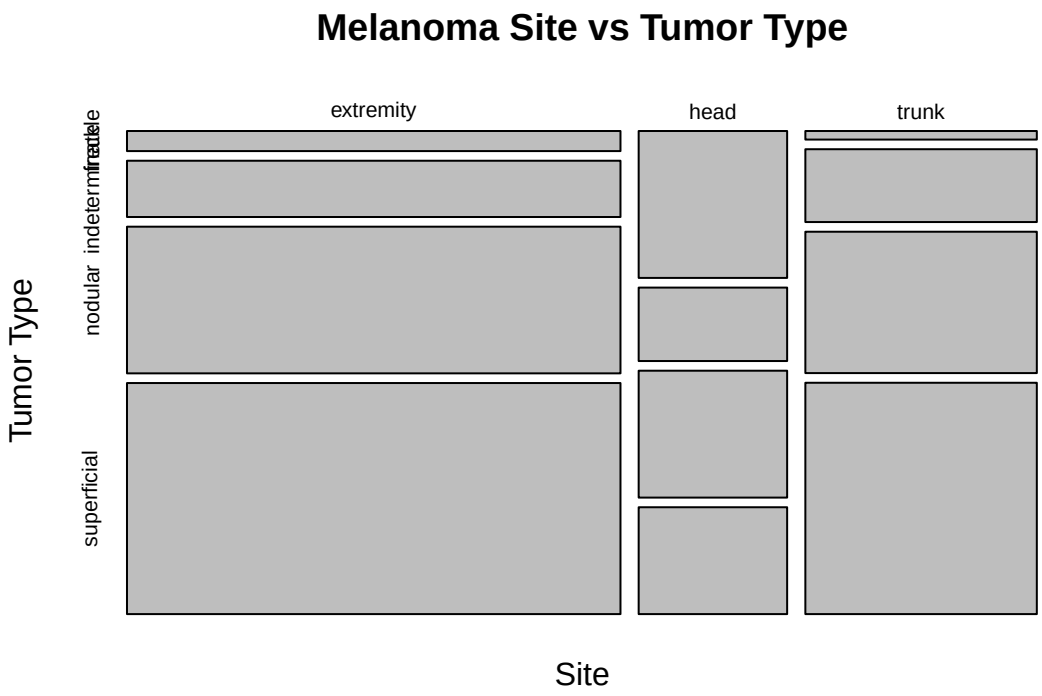
The residuals look more balanced after using the quasi-Poisson model, although there are still a few unusual points. The QQ plot is mostly close to a straight line, with some deviation at the ends.

```
data(melanoma)

tab <- xtabs(count ~ site + tumor, data = melanoma)
tab
```

```
##            tumor
## site        freckle indeterminate nodular superficial
##   extremity      10            28      73         115
##   head           22            11      19          16
##   trunk           2            17      33          54
```

```
mosaicplot(tab,
           main = "Melanoma Site vs Tumor Type",
           xlab = "Site",
           ylab = "Tumor Type")
```



The proportions of tumor types are different across the body sites, so site and tumor type do not appear to be independent.

```
chisq.test(tab)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tab
## X-squared = 65.813, df = 6, p-value = 2.943e-12
```

The chi-square test is significant x^2= 65.81, df = 6, $p < 0.001$, so site and tumor type are not independent.

```
mod2 <- glm(count ~ site + tumor,
            family = poisson,
            data = melanoma)

summary(mod2)
```

```
##
## Call:
## glm(formula = count ~ site + tumor, family = poisson, data = melanoma)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)          2.9554     0.1770  16.696  < 2e-16 ***
## sitehead            -1.2010     0.1383  -8.683  < 2e-16 ***
## sitetrunk           -0.7571     0.1177  -6.431 1.27e-10 ***
## tumorindeterminate   0.4990     0.2174   2.295   0.0217 *
## tumornodular         1.3020     0.1934   6.731 1.68e-11 ***
## tumorsuperficial     1.6940     0.1866   9.079  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 295.203  on 11  degrees of freedom
## Residual deviance:  51.795  on  6  degrees of freedom
## AIC: 122.91
##
## Number of Fisher Scoring iterations: 5
```

```
deviance(mod2)
```

```
## [1] 51.79501
```

```
df.residual(mod2)
```

```
## [1] 6
```

```
pchisq(deviance(mod2),
       df.residual(mod2),
       lower.tail = FALSE)
```

```
## [1] 2.050453e-09
```

The Poisson model does not fit the independence assumption well because the deviance is 51.80 with 6 degrees of freedom and the p-value is extremely small. This means site and tumor type are associated.

```
r <- residuals(mod2, type = "deviance")

res.tab <- xtabs(r ~ melanoma$site + melanoma$tumor)

res.tab
```

```
##              melanoma$tumor
## melanoma$site     freckle indeterminate     nodular superficial
##     extremity -2.31583297   -0.66016102  0.28104581  1.00813975
##     head       5.13537787    0.46798432 -0.49711084 -3.04533605
##     trunk     -2.82829426    0.54787007 -0.02173229  0.69899703
```

The largest residual is for freckle tumors on the head which is about 5.14, so there are more of these cases than the independence model expects. There are also large negative residuals for superficial tumors on the head and freckle tumors on the trunk, showing where the model fits poorly.

```r
mel.nohead <- droplevels(subset(melanoma, site != "head"))
tab.nohead <- xtabs(count ~ site + tumor, data = mel.nohead)
chisq.test(tab.nohead)
```

```
## Warning in chisq.test(tab.nohead): Chi-squared approximation may be incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  tab.nohead
## X-squared = 2.0254, df = 3, p-value = 0.5671
```

After removing the head cases, the chi-square test is not significant. This suggests that site and tumor type are independent for the remaining locations.